SDG BLOCKCHAIN ACCELERATOR

Market Penetration Strategy
Fuel Switch

1. Strategic Context and Market Entry Logic

Bosnia and Herzegovina sits in a structurally unique position within the European energy market. As a net exporter of electricity, the country's economic stability is directly tied to its ability to sell energy into neighboring European Union markets. However, the introduction of the Carbon Border Adjustment Mechanism (CBAM) in January 2026 fundamentally alters the conditions under which this trade can continue.

CBAM does not simply introduce a tax; it introduces a requirement for **verifiable proof of carbon intensity**. In the context of electricity exports, this translates into a binary condition: either energy can be proven to be renewable, or it is treated as carbon-intensive and taxed accordingly.

The core structural issue is that Bosnia and Herzegovina currently lacks the infrastructure to make this distinction at scale. Electricity from renewable and non-renewable sources is transmitted through the same grid, and there is no unified, operational system to certify origin.

This creates a market that is not emerging organically, but rather being **forced into existence by regulation**.

Fuel Switch is not entering a competitive SaaS market. It is entering a **time-constrained compliance gap**, where the first viable solution that achieves institutional acceptance can rapidly become the default.

The market penetration strategy is therefore built around a single principle:

Capture the compliance layer before regulatory enforcement makes adoption unavoidable.

2. Market Structure and Economic Opportunity

The starting point for understanding market penetration is quantifying the economic exposure.

Bosnia and Herzegovina exports approximately 15–20 TWh of electricity annually. A significant portion of this export is directed toward EU-linked markets, which will fall under CBAM requirements. Assuming a conservative 60% exposure, this results in approximately 10–12 TWh of electricity that must be compliant.

Within this volume, renewable energy is estimated to represent between 40% and 50%. This creates an immediately addressable certification volume of approximately 4–6 TWh annually, equivalent to 4–6 million MWh.

The pricing of Guarantees of Origin in European markets varies significantly depending on demand and regulatory pressure, but a conservative range of €1 to €5 per MWh provides a reasonable baseline.

This results in a total addressable market of:

- €4M to €30M annually in Bosnia and Herzegovina alone

However, this number significantly understates the opportunity.

CBAM is not limited to electricity. It extends to carbon-intensive industries such as steel and aluminum, both of which rely heavily on electricity inputs. As these sectors are progressively included, the demand for certified renewable energy, and therefore certificates, will increase structurally.

In addition, neighboring countries such as Serbia, Montenegro, and North Macedonia face identical regulatory exposure, with similar infrastructure gaps.

The realistic regional opportunity therefore expands into a €50M–€150M annual market over the medium term.

3. Market Dynamics and Urgency

What differentiates this market from traditional energy or SaaS markets is the nature of demand.

Demand is not driven by user preference, cost savings, or efficiency gains. It is driven by **regulatory necessity**.

Without certification:

- Renewable energy loses its economic advantage
- Exporters face direct financial penalties
- Traders cannot market energy as compliant

This creates three important dynamics:

First, adoption is **inevitable**. The question is not whether the market will adopt a solution, but which solution becomes embedded first.

Second, price sensitivity is initially low. In early stages, market participants prioritize **compliance over optimization**, allowing Fuel Switch to establish itself before pricing pressure increases.

Third, speed of execution becomes the primary competitive advantage. A technically superior solution that arrives late loses to a sufficiently credible solution that arrives early.

This is why the roadmap emphasis on early stakeholder engagement and pilot execution is critical.

4. Target Customer Strategy

Fuel Switch operates in a structurally interdependent market with two primary customer groups whose incentives are aligned but whose needs differ.

The first group consists of renewable energy producers. These include both large state-owned utilities and smaller private operators. Their core motivation is economic: they need to ensure that their energy is recognized as renewable in order to maintain pricing and market access.

However, these users are not technically sophisticated and do not want to manage certification complexity. Their ideal solution is one that abstracts away process complexity and operates as a managed service.

The second group consists of energy traders and offtakers. These entities sit at the interface between production and export markets. Their primary concern is compliance and auditability. They require certificates that can withstand regulatory scrutiny and be presented to EU counterparties.

The key insight is that adoption must occur at the level of **commercial relationships**, not individual users.

Fuel Switch is not selling to producers or traders independently. It is embedding itself into **producer–offtaker pairs**, where certification becomes part of the transaction itself.

This is why the pilot structure, linking PV Romanovci with ELIFA-Soil, is strategically sound. It validates the system within a real commercial flow rather than as a standalone tool.

5. Competitive Positioning

The competitive landscape is often misunderstood because it appears that existing registries solve the problem.

In reality, they do not.

Registry solutions such as Grexel operate at an institutional level. They are:

- slow to deploy
- fragmented across political entities
- not designed for commercial usability

Fuel Switch is not competing with registries. It is positioning itself as the **operational layer on top of them**.

This distinction is critical.

By remaining registry-agnostic, Fuel Switch avoids political resistance and instead becomes:

- an accelerator of existing systems
- a unifying interface across fragmented infrastructure
- a commercially usable layer for market participants

This positioning allows it to enter the market without requiring systemic replacement, which would be politically and operationally unfeasible.

6. Market Entry Strategy

The market entry strategy is deliberately structured in phases, each designed to remove a specific adoption barrier.

The first phase focuses on institutional alignment. In a market shaped by regulation, credibility must precede scale. Engagement with UNDP, MOFTER, and DERK is not a supporting activity; it is the foundation of market entry.

The objective is not formal approval, but **absence of resistance**. Achieving a “non-objection” position from regulators is sufficient to unlock the next phase.

The second phase is pilot commercialization. The goal is not technical validation, the platform is already functional, but commercial validation within a real transaction flow.

The success metric is not feature completeness, but whether a complete lifecycle, from issuance to retirement, can be demonstrated using real data and real counterparties.

The third phase focuses on volume expansion. At this stage, the strategy shifts from validation to coverage. The priority is not onboarding as many users as possible, but onboarding the **highest-volume producers**, maximizing certified MWh rather than user count.

The final phase is standardization. Once sufficient volume flows through the platform, Fuel Switch transitions from a solution to an implicit standard. At this point, adoption becomes self-reinforcing, as new participants adopt the system to remain compatible with existing market flows.

7. Revenue Model and Scaling Logic

The revenue model is intentionally aligned with energy volume rather than user count.

This is a critical design choice.

Charging per MWh ensures that:

- revenue scales with market activity
- incentives remain aligned with users
- barriers to entry remain low

Assuming an issuance fee in the range of €0.10 to €0.30 per MWh, and an initial onboarding of 1–2 TWh of certified energy within the first year, the platform can generate €100,000 to €600,000 in early-stage revenue.

As adoption expands to 5 TWh and beyond, revenue scales proportionally into the €500,000 to €1.5M range.

The long-term opportunity lies not in increasing pricing, but in increasing coverage.

8. Expansion Strategy

The expansion strategy follows a logical progression from local to regional dominance.

The initial focus is the Federation of Bosnia and Herzegovina, where the registry gap is most acute. This provides a clear entry point with minimal direct competition.

The next step is national integration, ensuring compatibility with Republika Srpska systems and positioning Fuel Switch as a unified interface.

Once a national footprint is established, the model can be replicated across the Western Balkans. The key advantage here is that CBAM creates identical regulatory conditions across multiple countries, allowing for rapid expansion using the same playbook.

9. Key Risks and Mitigation

The primary risk is regulatory inertia. Delays in engagement or lack of alignment with key ministries could slow adoption. This is mitigated through UNDP's involvement, which provides both access and credibility.

A second risk is data integration. Access to trusted meter data is essential for certification. The roadmap already accounts for this by including fallback mechanisms such as manually verified reports, ensuring that the pilot can proceed even without full API integration.

The third risk is user adoption. Producers may perceive the system as complex or burdensome. This is addressed by positioning Fuel Switch as a managed service, minimizing user involvement and emphasizing immediate financial benefits.

10. Strategic Conclusion

Fuel Switch is entering a market that will exist regardless of its presence. The question is not whether the market forms, but who captures it.

The combination of:

- regulatory urgency
- infrastructure gaps
- institutional backing
- and a service-based approach

creates a narrow but powerful window of opportunity.

If Fuel Switch succeeds in embedding itself within early transaction flows, it can transition from a solution provider to a **market standard**.

At that point, market penetration is no longer driven by sales, but by necessity.